chapter 2

Mechanisms of Hypertrophy

Increased muscle protein accretion following resistance exercise has been attributed to three primary mechanisms: mechanical tension, metabolic stress, and muscle damage (240). This chapter addresses each of these mechanisms and the theoretical rationale for their promotion of a hypertrophic response.

Mechanical Tension

Skeletal muscle is highly responsive to alterations in mechanical loading. Accordingly, a number of researchers have surmised that *mechanical tension* is the primary driving force in the hypertrophic response to regimented resistance training (77, 88) and at the very least initiates critical hypertrophy-related intracellular signaling following resistance exercise (226). In simple terms, mechanical tension can be defined as a force normalized to the area over which it acts, with units expressed in either newtons per square meter or pascals (31). Mechanical tension alone has been shown to directly stimulate mTOR (113), possibly through activation of the extracellular signal–regulated kinase/tuberous sclerosis complex 2 (ERK/TSC2) pathway (188). It is theorized that these actions are mediated via the synthesis of the lipid second messenger phosphatidic acid by phospholipase D (113, 206). There also is evidence that phosphatidic acid can phosphorylate $p70^{S6K}$ independent of mTOR (151), presenting another potential avenue whereby mechanical stimuli may directly influence muscle protein synthesis.

Research indicates that mechanosensors are sensitive to both the magnitude and temporal aspects of loading. Using an *in situ* model (i.e., examining an intact muscle within the animal), Martineau and Gardiner (167) subjected rat plantaris muscles to peak concentric, eccentric, isometric, and passive tensions. Results showed tension-dependent phosphorylation of c-Jun N-terminal kinase (JNK) and ERK1/2; eccentric actions generated the greatest effect, and passive stretch generated the least. Peak tension was determined to be a better predictor of mitogen-activated protein kinase (MAPK) phosphorylation than either time under tension or rate of tension development. In a follow-up study by the same lab (168), an in situ evaluation of the rat gastrocnemius muscle showed a linear relationship between time under tension and the signaling of JNK, whereas the rate of change of tension showed no effect. This suggests that time under tension is an important parameter for muscle hypertrophic adaptations. In support of these findings, Nader and Esser (193) reported increased activation of $p70^{S6K}$ following both high-intensity and low-intensity electrical stimuli of the rat hind limb; however, the response was not as prolonged following the low-intensity protocol. Similarly, in vitro research shows a magnitude-dependent effect on $p70^{S6K}$ signaling when mouse C2C12 myoblasts are subjected to biaxial strain (74).

Mechanosensors also appear to be sensitive to the type of load imposed on muscle tissue. Stretch-induced mechanical loading elicits the deposition of sarcomeres longitudinally (i.e., in series), whereas dynamic muscular actions increase cross-sectional area in parallel with the axes (74). Moreover, the hypertrophic response can vary based on the type of muscle action. Isometric and eccentric actions stimulate the expression of distinct genes in a manner that cannot be explained by differences in the magnitude of applied mechanical force (74). These examples highlight the intricate complexity of mechanosensors and their capacity to distinguish between types of mechanical information to produce an adaptive response. What follows is a discussion of how mechanical forces regulate muscle hypertrophy via mechanotransduction and associated intracellular signaling pathways.

KEY POINT

Mechanical tension is the most important factor in training-induced muscle hypertrophy. Mechanosensors are sensitive to both the magnitude and the duration of loading, and these stimuli can directly mediate intracellular signaling to bring about hypertrophic adaptations.

Mechanotransduction

Exercise has a profound effect on muscle protein balance. When muscles are mechanically overloaded and then provided with appropriate nutrients and recovery, the body initiates an adaptive response that results in the accretion of muscle proteins. Transmission of mechanical forces from the sarcomeres to tendons and bones occurs both longitudinally along the length of the fiber and laterally through the matrix of fascia tissue (259). The associated response is accomplished through a phenomenon called *mechanotransduction*, whereby mechanical forces in muscle are converted into molecular events that mediate intracellular anabolic and catabolic pathways (see figure 2.1) (308).

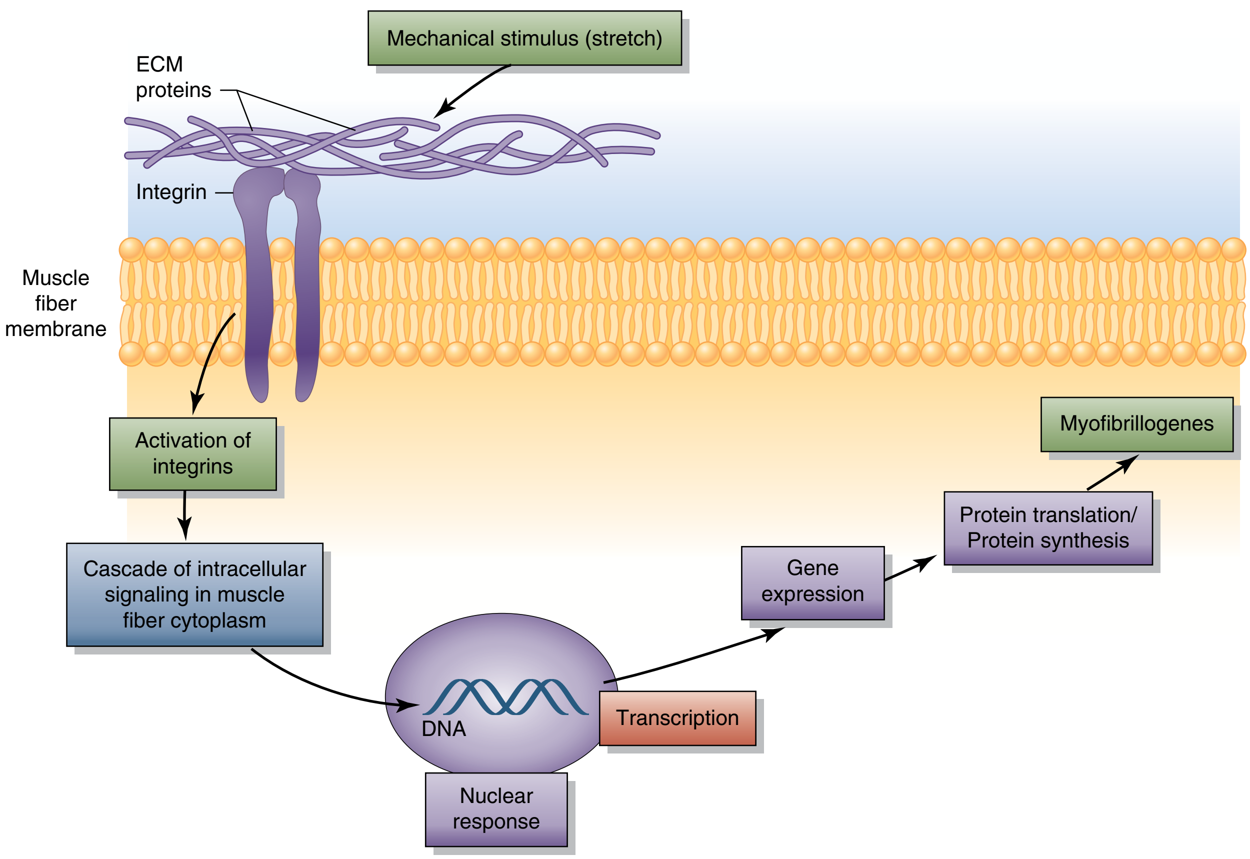

FIGURE 2.1 The process of mechanotransduction.

Adapted from P.G. De Deyne, "Application of Passive Stretch and Its Implications for Muscle Fibers," *Physical Therapy* 81, no. 2 (2001): 819-827.

A diverse array of tissue and substances help to carry out mechanotransduction, including stretch-activated ion channels, caveolae, integrins, cadherins, growth factor receptors, myosin motors, cytoskeletal proteins, nuclei, and the extracellular matrix (74). These mechanosensory elements do not function independently, but rather act in a coordinated manner with structural components such as the cytoskeleton to elicit intracellular events (74). Central to the process are mechanosensors that detect mechanical tension and transduce the stimuli into chemical signals within the myofiber. Integrins have been identified as a primary mechanosensor. These receptors reside at the cell surface and interact with the extracellular matrix to facilitate the transmission of mechanical and chemical information from the outside to the inside of the cell (307, 308). Integrins mediate intracellular signal transduction as part of focal adhesion complexes (i.e., costameres), which are sarcolemmal proteins that bridge the connection between the extracellular matrix and the cytoskeleton. Focal adhesion complexes can directly enhance protein translation via activation of ribosomal proteins, and their disruption impairs intracellular anabolic signaling (173). Emerging evidence shows that an enzyme called focal adhesion kinase (FAK) serves as a key player in signal initiation (48). The expression of FAK displays load-dependent characteristics whereby its activation is suppressed during unloading and heightened during mechanical overload, highlighting the mechanosensitive role of FAK in exercise-induced hypertrophy (9).

Other stimuli and sensors have been surmised to play a role in hypertrophic adaptations. For example, emerging evidence implicates titin as a primary mechanosensor, and the level of signaling depends on its passive stiffness: High stiffness mediates a stronger anabolic response whereas low stiffness moderates the response (285). Moreover, G protein–coupled receptors, which show structural similarity to integrin receptors, are proposed as a potential link between mechanical force transduction and upregulation of intracellular anabolic pathways (301). It also has been hypothesized that the "flattening" of myonuclei during mechanical loading may act as a sensory signal for various growth-related proteins (e.g., YAP) to translocate from the cytosol to the nucleus and thus initiate anabolism (294), although this theory remains speculative. Overall, our understanding of the stimuli and sensors involved in mechanotransduction is poorly characterized; the topic will be an important area of future research.

Once forces are transduced, intracellular enzymatic cascades carry out signaling to downstream targets that ultimately shift muscle protein balance to favor synthesis over degradation. Certain pathways act in a permissive role, whereas others directly mediate cellular processes that influence mRNA translation and myofiber growth (172). A number of primary anabolic signaling pathways have been identified, including the PI3K/Akt pathway, MAPK pathways, calcium-dependent pathways, and the phosphatidic acid pathway (see figure 2.2), among others. Although these pathways may overlap at key regulatory steps, there is evidence they may be interactive rather than redundant (276).

Alternatively, muscle catabolism is regulated by four proteolytic systems: autophagy-lysosomal, calcium-dependent calpains, the cysteine protease caspase enzymes, and the ubiquitin–proteasome system (211). The 5′-AMP-activated protein kinase (AMPK) pathway is believed to act as a metabolic master switch in these systems. It is activated in response to environmental stressors (e.g., exercise) to restore cellular energy balance via an increase of catabolic processes and a suppression of anabolic processes (see figure 2.3 on page 34). The MSTN-SMAD pathway also is considered a strong catabolic regulator of muscle protein accretion.

Signaling Pathways

This section provides a general overview of the primary anabolic intracellular signaling pathways and their significance to skeletal muscle hypertrophy. Although huge strides have been made to elucidate these pathways, our understanding of their relative importance is limited at this time.

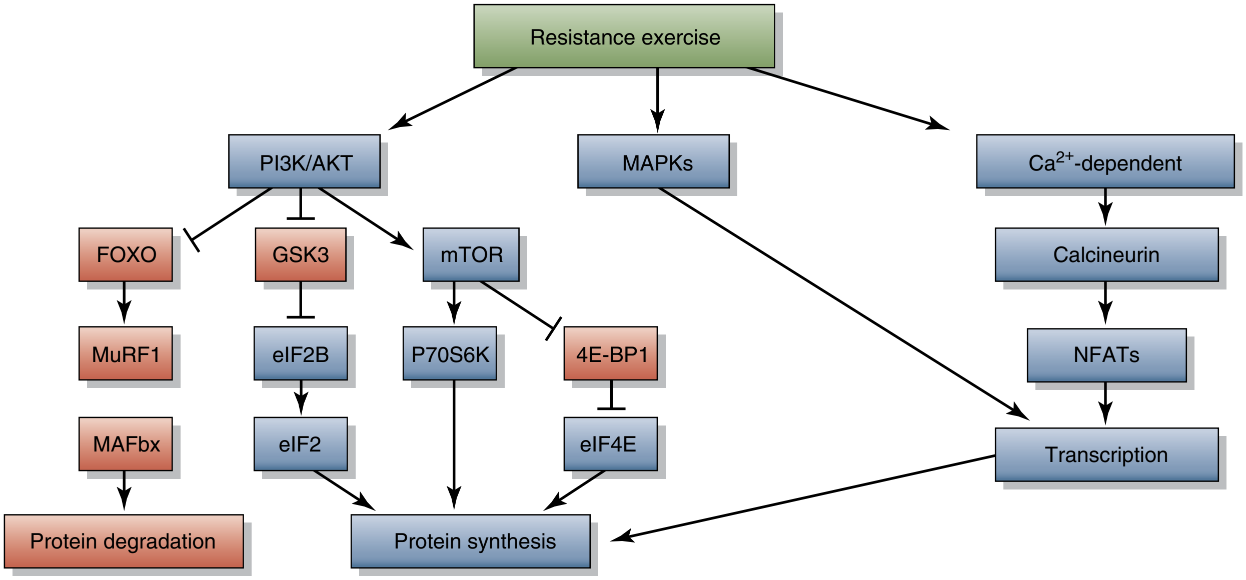

FIGURE 2.2 Primary anabolic intracellular signaling pathways.

Reprinted from B.J. Schoenfeld, "Potential Mechanisms for a Role of Metabolic Stress in Hypertrophic Adaptations to Resistance Training," *Sports Medicine* 43, no. 3 (2013): 179-194, by permission of Springer Nature.

KEY POINT

Numerous intracellular signaling pathways have been identified in skeletal muscle including PI3K/Akt, MAPK, phosphatidic acid, AMPK, and calcium-dependent pathways. The serine/threonine kinase mTOR has been shown to be critical to resistance training–induced hypertrophic adaptation.

PI3K/Akt Pathway

The phosphatidylinositol 3-kinase (PI3K)/Akt pathway is considered a master network for regulating skeletal muscle growth (20, 125, 274). Akt, also known as protein kinase B (PKB), acts as a molecular upstream nodal point that functions both as an effector of anabolic signaling and a dominant inhibitor of catabolic signals (279). Multiple isoforms of Akt have been identified in skeletal muscle (Akt1, Akt2, Akt3), and each has a distinct physiological role. Of these isoforms, Akt1 appears to be most responsive to mechanical stimuli (307). Early research indicated that high mechanical intensities were required to activate Akt; however, subsequent studies demonstrate evidence to the contrary (307).

A primary means by which Akt carries out its actions is by signaling mTOR, which has been shown to be critical to hypertrophic adaptations induced by mechanical loading. mTOR, named because the pharmacological agent rapamycin antagonizes its growth-promoting effects, exists in two functionally distinct signaling complexes: mTORC1 and mTORC2. Only mTORC1 is inhibited by rapamycin (203), and this complex was originally thought to be responsible for mTOR's hypertrophic regulatory actions; however, recent research indicates that rapamycin-insensitive mTORC2 also plays a role in load-induced anabolism (202). Some evidence shows that early increases in muscle protein synthesis are regulated by mTORC1, while continued elevations at later time points involve rapamycin-insensitive or perhaps even mTOR-independent mechanisms (92). It should be noted that mTOR is regulated by a variety of inputs and functions as an energy and nutrient sensor: Elevated energy levels promote its activation while reduction in energy levels and nutrient availability result in its suppression (19).

Once activated, mTOR exerts its effects by turning on various downstream anabolic

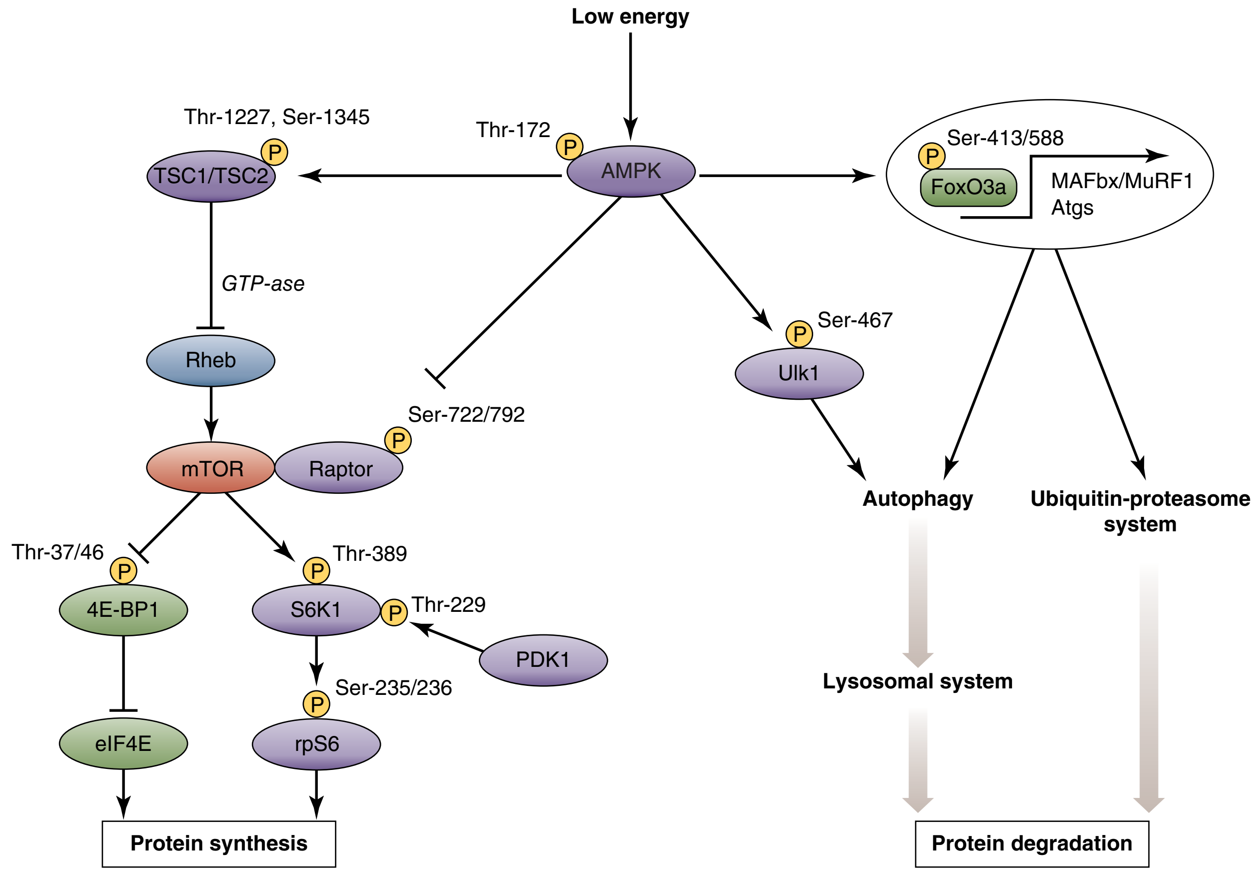

FIGURE 2.3 Primary proteolytic pathways.

Reprinted by permission from A.M.J. Sanches et al., "The Role of AMP-Activated Protein Kinase in the Coordination of Skeletal Muscle Turnover and Energy Homeostasis," *American Journal of Physiology—Cell Physiology* 303, no. 5 (2012): C475-C485.

effectors. A primary target of mTOR is $p70^{S6K}$, which plays an important role in the initiation of mRNA translation (90). mTOR also exerts anabolic effects by inhibiting eukaryotic initiation factor 4E-binding protein 1 (eIF4EB1), a negative regulator of the eIF4E protein that is a potent mediator of protein translation (86). Interestingly, persistently elevated basal levels of mTOR have been shown to impair fast-twitch fiber growth in mice and contribute to anabolic resistance in elderly humans; thus, it has been postulated that mTOR mediates hypertrophy within a given range, and deviations outside of this range may be detrimental to the growth process (56, 92).

Signaling through PI3K/Akt also regulates mTOR-independent growth regulatory molecules to directly inhibit catabolic processes. For one, Akt phosphorylates FOXO proteins—a subgroup of the Forkhead family of transcription factors that encourage atrophy—which induces their translocation from the nucleus to the cytoplasm (90, 106). The cytoplasmic sequestration of FOXO proteins, in turn, blocks upregulation of the ubiquitin ligases MuRF-1 and atrogin-1 (also called MAFbx) and thus helps to lessen the extent of muscle protein breakdown. Indeed, activation of Akt was found to be sufficient to impair increases in the atrophy-associated enzymes MuRF-1 and atrogin-1 transcription via FOXO phosphorylation (86). Akt also suppresses the activation of glycogen synthase kinase 3 beta (GSK3β), which blocks protein translation initiated by the eIF2B protein (86, 206). As opposed to mTORC1, which regulates the translation of a small subset of mRNAs, eIF2B is believed to control the translation initiation of virtually all mRNAs, and therefore acts to regulate global rates of protein synthesis (90). Thus, the anticatabolic actions of PI3K/Akt may indirectly provide an even more potent stimulus for growth than its anabolic effects.

The hypertrophic properties of PI3K/Akt are incontrovertible. Induction of the pathway has been shown to mediate protein translation both in vitro and in vivo, as well as promote myoblast differentiation (86). However, recent research indicates that PI3K/Akt activation is not obligatory for increases in muscle hypertrophy (292). Resistance exercise activates $p70^{S6K}$ in humans via an Akt-independent pathway (60, 170, 271). Moreover, mTOR can be activated via a variety of intracellular signals other than PI3K/Akt, indicating that the pathways influencing growth are complex and diverse. The primary anabolic property of Akt1 during load-induced hypertrophy may be in its ability to regulate satellite cell proliferation (189).

MAPK Pathways

Mitogen-activated protein kinase (MAPK) is a primary regulator of gene expression, redox status, and metabolism (141). With respect to exercise-induced muscle growth, MAPK is believed to link cellular stress with an adaptive response in myofibers, modulating their growth and differentiation (231). It is theorized that the maximal anabolic response to resistance exercise is at least in part reliant on coactivation of the MAPK and mTORC1 signaling cascades (210). Moreover, MAPK has been implicated in the regulation of ribosome biogenesis, which is critical to sustained increases in muscle growth (67). Three distinct MAPK signaling modules are associated with mechanically stimulated hypertrophic adaptations: ERK1/2, p38 MAPK, and JNK. Activation of these modules depends on the type, duration, and intensity of the stimulus.

ERK1/2 is upregulated by both aerobic endurance and resistance training, and the magnitude of its phosphorylation correlates with the intensity of exercise (141). Studies investigating the role of ERK1/2 in the regulation of muscle mass have been somewhat conflicting. On one hand, there is evidence that it mediates satellite proliferation and induces muscle protein synthesis (85); on the other hand, some research shows opposite effects (61). That said, early signaling of mTORC1 likely occurs through activation of the ERK/TSC2 pathway (188). Whereas Akt and ERK1/2 both stimulate mTOR to a similar extent, their combined effects lead to an even greater stimulation compared to either alone (305). Moreover, the two pathways appear to be synergistic to satellite cell function; ERK1/2 stimulates cell proliferation, and PI3K facilitates differentiation (101).

Activation of p38 MAPK occurs primarily following aerobic endurance exercise. Four p38 isoforms have been identified (p38α, p38β, p38δ, and p38γ). Of these isoforms, p38γ is specific to muscle tissue, whereas p38α and p38β are expressed throughout the body; p38δ does not appear to be involved with muscular actions; p38γ is preferentially upregulated in slow-twitch fibers while remaining largely inactive in fast-twitch fibers (73). Moreover, a loss of p38γ in rat and mouse models is associated with a decrease in slow-twitch fiber size and no change in fast-twitch fibers (73). As opposed to directly binding to DNA, p38 MAPK mediates transcription of target genes by activating other transcription factors (96). There is evidence that p38 may regulate hypertrophy by stimulating Notch signaling, which has been deemed essential for the activation, proliferation, and progression of myogenic satellite cells necessary for muscle regeneration and repair (25).

Of all the MAPK modules, JNK appears to be the most sensitive to mechanical tension, and it is particularly responsive to eccentric actions. Contraction-induced phosphorylation of JNK correlates with a rapid rise in mRNA of transcription factors that mediate cell proliferation and DNA repair (7, 8), indicating a role in muscle regeneration following intense exercise. Moreover, JNK phosphorylation displays a linear increase, with heightened levels of contractile force (141). However, the specific role of JNK in exercise-induced muscle hypertrophy remains undetermined. Some researchers have deemed JNK a molecular switch that, when activated, stimulates a hypertrophic response and, when suppressed, induces smaller, more oxidative muscle fibers; these effects were found to be regulated, at least in part, by myostatin inhibition (152). However, other research suggests that inhibition of JNK actually enhances muscle protein accretion (25), so its precise role in muscle anabolism remains somewhat unclear.

The interplay between the MAPK modules and their potential hypertrophic synergism with one another has yet to be established. In response to synergist ablation of the rat gastrocnemius, p38α MAPK phosphorylation occurred early following overload and remained elevated in both slow-twitch soleus and fast-twitch plantaris muscles over the ensuing 24-hour study period. Conversely, ERK2 and JNK phosphorylation increased transiently post-ablation; levels returned to that of sham-operated controls (placebo-controlled surgical interventions) by 24 hours. The implications of these findings are not clear at present.

Calcium-Dependent Pathways

Intracellular calcium plays an important role in signal transduction in a variety of cell types, including skeletal muscle (38). An increase in myoelectrical activity substantially elevates calcium levels within myofibers, and this alteration is considered to be a primary mediator of skeletal muscle gene expression (38). Increased intracellular calcium levels have been shown to amplify protein synthesis via TORC1 signaling, although the mechanism of action is as yet unknown (173). Moreover, elevations in extracellular ATP promote muscle hypertrophy via an increase in intracellular calcium levels, leading to subsequent downstream anabolic signaling in rodents (124). Intriguingly, the hypertrophy occurred in the soleus but not the plantaris muscles, suggesting that calcium-dependent effects are specific to slow-twitch fibers.

Various calcium-dependent pathways have been implicated in the control of skeletal muscle mass. Calcineurin, a calcium-regulated phosphatase, is believed to have a particularly important role in muscular adaptations. Calcineurin is activated by a sustained increase in intracellular calcium levels. Once aroused, it acts on various downstream anabolic effectors, including myocyte-enhancing factor 2 (MEF2), GATA transcription factors, and nuclear factor of activated T cells (NFAT) (183). Calcineurin has been shown to promote hypertrophy in all fiber types, whereas its inhibition prevents growth even when muscles were subjected to overload (57, 58). Early evidence suggested that, along with PI3K/Akt signaling, activation of calcineurin was required for IGF-1–mediated hypertrophic adaptations (119). It was hypothesized that these effects were expressed via activation of NFAT, which in turn mediated the signaling of transcriptional regulators such as proliferator-activated receptor gamma coactivator 1-alpha (PGC1α) and striated muscle activator of Rho signaling (STARS) (162, 169). However, subsequent research challenged these findings, indicating that calcineurin in muscle was primarily responsible for producing a shift toward a slower phenotype (195, 267). When considering the body of literature as a whole, evidence suggests both correlative and causal links between calcineurin and muscle fiber size, especially in slow-twitch fibers (119). That said, muscle growth does not appear to be dependent on calcineurin activity (14), and the role (if any) that the enzyme plays in the hypertrophic response to exercise overload is unclear.

The calcium-calmodulin-dependent kinases (i.e., CaMKII and CaMKIV) also have a prominent role in muscle plasticity. CaMKII and CaMKIV have multiple isoforms that detect and respond to calcium signals via multiple downstream targets (38). CaMKII is activated by both acute and long-duration exercise, indicating that it mediates muscle growth as well as mitochondrial biogenesis (38). Interestingly, increases in one of the CaMKII isoforms (CaMKIIγ) occurs during muscle atrophy, leading to the possibility that it is upregulated as a compensatory response to counter the wasting process (38).

Phosphatidic Acid Pathway

Phosphatidic acid (PA) is a lipid second messenger that regulates a diverse array of cellular processes, including muscle growth in response to mechanical load. The activation of PA is mediated via several classes of enzymes. In particular, it is synthesized by phospholipase D1 (PLD1), which hydrolyzes phosphatidylcholine into PA and choline. Once activated, PA exerts effects on both protein synthesis and proteolysis. This is principally accomplished by its direct binding to mTOR and then activating $p70^{S6K}$ activity (248, 307). PA also can phosphorylate $p70^{S6K}$ in an mTOR-independent manner, presenting yet another path whereby mechanical stimuli may directly drive anabolic processes (151). In addition, overexpression of PLD1 is associated with a decrease in catabolic factors such as FOXO3, atrogin-1, and MuRF-1 (91). Suppression of these atrophy-related genes is believed to be due to Akt phosphorylation and subsequent activation of mTORC2. Thus, PLD1 carries out anabolic and anticatabolic actions through varied intracellular mechanisms.

PA is highly sensitive to mechanical stimulation. Both ex vivo passive stretch (i.e., stretch performed on a muscle removed from the body) and in vivo eccentric actions (i.e., actions of a muscle that is intact in the body) were found to increase PA and mTOR signaling (91). Moreover, administration of 1-butanol—a PLD antagonist—blunts both PA synthesis and mTOR signaling (114). In combination, these data indicate that PLD-derived PA is integrally involved in the mechanical activation of mTOR (91). It should be noted that PA can be synthesized by alternative enzymes, and there is some evidence that its activation by diacylglycerol kinase may play a role in its hypertrophic effects as well.

AMPK Pathway

The trimeric enzyme 5′-AMP-activated protein kinase (AMPK) plays a key role in the regulation of cellular energy homeostasis. AMPK acts as a cellular energy sensor; its activation is stimulated by an increase in the AMP/ATP ratio (90). As such, conditions that elicit substantial intracellular energy stress—including exercise—can activate AMPK. Once activated, AMPK suppresses energy-intensive anabolic processes such as protein synthesis and amplifies catabolic processes, including protein breakdown (90).

Because of its inherent actions, AMPK is theorized to be involved in the maintenance of skeletal muscle mass. This contention is supported by evidence showing that knockout (inactivation) of AMPK in animal models causes hypertrophy both in vitro and in vivo (90). Alternatively, activation of AMPK by AICAR—an AMPK agonist—promotes myotube atrophy, whereas its suppression counteracts the atrophic response (90). Taken together, these findings indicate that AMPK impairs muscle hypertrophy by suppressing protein synthesis and stimulating proteolysis.

The precise mechanisms by which AMPK carries out its actions are still being elucidated. Proteolytic effects of AMPK appear to be related at least in part to its influence over atrogin-1. Protein degradation induced by AMPK agonists (AICAR and metformin) has been found to correlate with atrogin-1 expression, whereas another AMPK antagonist (Compound C) blocks such expression. Evidence shows that these actions may involve an AMPK-induced increase in FOXO transcription factors, thereby stimulating myofibrillar protein degradation via atrogin-1 expression (194). AMPK has also been shown to induce protein degradation via activation of *autophagy* (regulated cell degradation by organelles termed lysosomes) (90), although it remains to be determined whether this mechanism plays a role in skeletal muscle adaptations following mechanical overload. Other research indicates that AMPK reduces cell differentiation of myoblasts and thus negatively affects hypertrophic adaptations without necessarily accelerating protein degradation (286).

In addition to the catabolic actions of AMPK, compelling evidence suggests that it suppresses the rate of protein synthesis. It is theorized that this negative influence is mediated at least in part by antagonizing the anabolic effects of mTOR, either by direct phosphorylation of mTOR, indirect phosphorylation of the tuberous sclerosis complex (TSC), or both, which has the effect of inhibiting the Ras homolog enriched in brain (RHEB) (187, 258). The upshot is an inhibition of translation initiation (19), the rate-limiting step in muscle protein synthesis.

Another potential means whereby AMPK is theorized to negatively affect muscle protein synthesis is the inhibition of translation elongation and the indirect suppression of the anabolic effector eIF3F (90). Thus, there are multiple potential mechanisms for AMPK-mediated regulation of protein synthesis.

A number of studies lend support to the theory that AMPK plays a role in the muscular adaptations in response to regimented exercise training. AMPK activation shows a strong inverse correlation with the magnitude of muscle hypertrophy following chronic overload (275). In addition, AMPK inhibition is associated with an accelerated growth response to mechanical overload, whereas its activation attenuates hypertrophy (90). However, other research calls into question the extent to which AMPK regulates exercise-induced hypertrophy. In humans, mTOR signaling and muscle protein synthetic rate are elevated following resistance exercise despite concomitant activation of AMPK (54). This indicates that, at the very least, the activation of AMPK is not sufficient to completely blunt growth. Moreover, growth in mice lacking the primary upstream kinase for AMPK was not enhanced following functional overload, casting uncertainty about the importance of AMPK in muscular adaptations to mechanical loading (175).

MSTN-SMAD Pathway

The role of MSTN in muscle hypertrophy was outlined in chapter 1 and thus will only be briefly discussed here. MSTN, a member of the transforming growth factor-β superfamily, is a potent negative regulator of muscle growth. Knockout of the MSTN gene causes hypermuscularity whereas its overexpression causes atrophy. MSTN carries out its effects through activation of SMAD2 and SMAD3 (via phosphorylation of activin Type I receptors), which in turn translocate to the cell nucleus and regulate transcription of target genes via interaction with DNA and other nuclear factors (229).

MSTN plays an important role in the maintenance of muscle mass, and its expression decreases in almost all resistance exercise studies. Intriguingly, some research shows correlations between resistance training–induced reductions in MSTN and subsequent increases in muscle growth (222), while other research has failed to demonstrate such associations (66). The specific role of MSTN with respect to its hypertrophic effects during resistance training therefore remains to be fully elucidated. For further insights on the topic, please see the myokine section that covers MSTN in chapter 1.

Metabolic Stress

Although the importance of mechanical tension in promoting muscle growth is indisputable, there is evidence that other factors also play a role in the hypertrophic process. One such factor proposed to be of particular relevance to exercise-induced anabolism is *metabolic stress* (230, 244, 255). Simply stated, metabolic stress is an exercise-induced accumulation of metabolites, particularly lactate, inorganic phosphate, and H+ (261, 272). However, it should be noted that approximately 4,000 metabolites have been detected in human serum (294), and thus other metabolic byproducts may be relevant to training-related adaptations as well. Several researchers have surmised that metabolite buildup may have an even greater impact on muscle hypertrophy than high-force development (250), although other investigators dispute this assertion (71).

Metabolic stress is maximized during exercise that relies heavily on anaerobic glycolysis for energy production, which is characterized by a reduced PCr concentration, elevated lactate levels, and a low pH. Anaerobic glycolysis is dominant during exercise lasting 15 to 120 seconds, and corresponding metabolite accumulation causes peripherally (as opposed to centrally) induced fatigue (i.e., fatigue related to metabolic or biochemical changes, or both, as opposed to reductions in neural drive) (227). Research shows that performing 1 set of 12 repetitions to failure (with a total time under tension of 34 to 40 seconds) elevates muscle lactate levels to 91 mmol/kg (dry weight), and

KEY POINT

Evidence suggests that metabolic stress associated with resistance training can promote increases in muscle hypertrophy, although it is unclear whether these effects have a synergistic relationship with mechanical tension or are redundant.

values increase to 118 mmol/kg after 3 sets (160). In contrast, minimal metabolite buildup is seen in protocols involving very heavy loading (≥90% of 1RM) because the short training durations involved (generally <10 seconds per set) primarily tap the phosphagen system for energy provision. In addition, muscle oxygenation is compromised during resistance training that relies on fast glycolysis. The persistent compression of circulatory flow throughout a longer-duration set results in acute hypoxia, thereby heightening metabolite buildup (268). The combination of these factors causes the rapid accumulation of intramuscular metabolites along with a concomitant decrease in pH levels (260).

Typical bodybuilding routines are intended to capitalize on the growth-promoting effects of metabolic stress at the expense of higher intensities of load (77, 240). These routines, which involve performing multiple sets of 8 to 12 repetitions per set with relatively short interset rest intervals (145), have been found to increase metabolic stress to a greater degree than higher-intensity regimens typically employed by powerlifters (135-137). It is well documented that despite regular training at moderate intensities of load, bodybuilders display hypermuscular physiques and levels of lean body mass at least as great as, if not greater than, those achieved by powerlifters (77, 128). Indeed, there is evidence that bodybuilding-type routines produce superior hypertrophic increases compared to higher-load powerlifting-style routines (39, 171, 238), although findings are not consistent across all trials when equating for volume load (32, 243).

Evidence suggests that various metabolites can directly function as hypertrophic stimuli. Of particular relevance, several studies have demonstrated that culturing muscle cells in vitro with lactate enhances anabolic signaling and myogenesis (204, 205, 282, 303), and similar results have been shown using an in vivo murine model (34). An in vivo study by Oishi and colleagues (205) showed that daily oral administration of a lactate and caffeine supplement combined with treadmill exercise produced significantly greater hypertrophic increases in the gastrocnemius and tibialis anterior muscles of male rats than in the exercise-only and sedentary control groups. The inclusion of caffeine in the supplement confounds these findings, although acute results from the same study showed only minor benefits of the combination of lactate and caffeine versus lactate alone on markers of intracellular anabolic signaling. Similarly, Ohno and colleagues (204) found that oral lactate administration in mice increased fiber cross-sectional area of the tibialis anterior along with a corresponding increase in Pax7-positive nuclei in the muscle. Most recently, Tsukamoto and colleagues (282) demonstrated that intraperitoneal injection of lactate in rodents, at levels similar to those seen following resistive exercise, elicited significantly greater hypertrophy of the tibialis anterior muscle compared to saline injection. Although the mechanisms of potential hypertrophic effects of lactate have not been elucidated, it is hypothesized that they may be regulated via calcium-dependent signaling pathways (205). Notably, these animal studies do not replicate the in vivo response during human exercise, so the practical implications of results must be interpreted cautiously. Intriguingly, lactate production serves to inhibit activity of histone deacetylase (149), a negative regulator of muscle growth (179), thereby providing another potential avenue for inducing hypertrophic effects.

There also is evidence that H+ can alter hypertrophic adaptations. Type II fibers are particularly sensitive to acidosis. It is theorized that the intramuscular buildup of H+ impairs calcium binding in these fibers, causing a progressive reduction in their force-producing capacity as metabolically taxing exercise continues (97). Accordingly, this places an increased burden on Type I fibers to maintain force output and conceivably enhance their development. Several additional factors are theorized to mediate hypertrophic adaptations from exercise-induced metabolic stress, including increased fiber recruitment, myokine production alterations, cell swelling, metabolite accumulation, and elevated systemic hormone production (93, 94, 198, 265). What follows is a discussion of how these factors are thought to drive anabolism (figure 2.4).

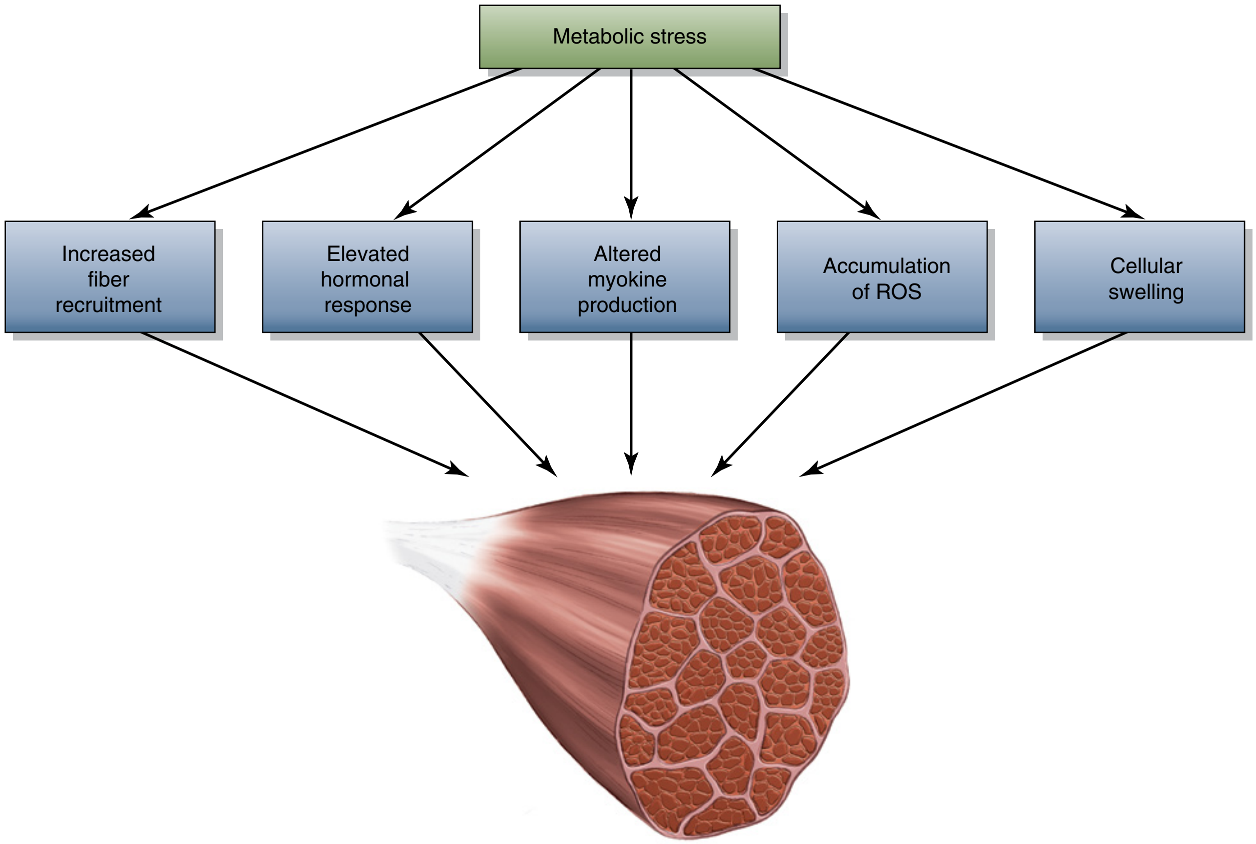

FIGURE 2.4 Mechanisms of metabolic stress.

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Fiber Recruitment

As discussed in chapter 1, muscle fiber recruitment is carried out in an orderly fashion whereby low-threshold motor units are recruited first and then higher-threshold motor units are progressively recruited thereafter to sustain muscle contraction depending on force demands (110). Although heavy loading rapidly activates the full spectrum of fiber types, research indicates that metabolic stress increases the recruitment of higher-threshold motor units even when lifting light loads. Studies show that as fatigue increases during sustained submaximal exercise, recruitment thresholds correspondingly decrease (116, 234, 293). Accordingly, activation of fast-twitch fibers is high provided a set is carried out to the point of muscular failure. Studies employing electromyography (EMG) (265, 266), glycogen depletion (123), and organic phosphate splitting (260, 261) have all demonstrated increased fast-twitch fiber activation in BFR training, causing some researchers to speculate that this is the primary factor by which occlusion mediates anabolism (155, 182).

The precise mechanisms whereby metabolic stress augments fast-twitch fiber recruitment are not entirely clear. It has been hypothesized that H+ accumulation plays a substantial role by inhibiting contractility in working fibers and thus promoting the recruitment of additional high-threshold motor units (51, 186, 266). MacDougall and colleagues (160) proposed that fatigue during single-set training to failure is due to a combination of acidosis and PCr depletion, whereas acidosis is more likely the cause in multiset resistance exercise.

Although it would seem that increased fiber recruitment is at least partly responsible for the

RESEARCH FINDINGS

BLOOD FLOW RESTRICTION

The impact of metabolic stress on hypertrophic adaptations is exemplified by *blood flow restriction* (BFR) training studies. BFR training involves restricting venous inflow via the use of a pressure cuff while training (figure 2.5) with light weights (generally equating to <40% of 1RM), thereby heightening ischemia in the muscle as it contracts.

The prevailing body of literature shows that BFR training stimulates anabolic signaling and muscle protein synthesis (78) and markedly increases muscle growth (156) despite employing loads often considered too low to promote significant hypertrophy (32, 140).

FIGURE 2.5 A blood flow restriction implement on an arm.

It has been speculated that metabolic stress is the driving force behind BFR-induced muscle hypertrophy. Significant metabolite buildup has been noted during such training (154), pointing to an association between metabolic stress and muscle growth. In further support of this contention, significant increases in cross-sectional area of the thigh muscle were found in college-aged males after 3 weeks of walking with BFR of the legs (3). Given that healthy young subjects generally do not gain muscle from performing low-intensity aerobic exercise, the study provides strong evidence that factors other than mechanical tension were responsible for hypertrophic adaptations. Indeed, increases in muscle cross-sectional area were found to be significantly correlated with the changes in inorganic phosphate (r = .876) and intramuscular pH (r = .601) during BFR training carried out at 20% of 1RM. Results suggest that metabolic stress generated during exercise may be a key regulator of muscle growth (263). In further support of this hypothesis, evidence indicates that ischemia—a potent mediator of metabolic stress—upregulates anabolic myokines during exercise performance, although the duration of these elevations is rather transient, returning to baseline values within approximately 30 minutes after cessation of training (249).

Studies investigating resistance training under conditions of hypoxia provide further evidence of an association between metabolic stress and muscle growth. Kon and colleagues (133) found that breathing 13% oxygen during a multiset, low-load (~50% of 1RM) protocol with fairly short interset rest intervals (~1 minute) significantly heightened blood lactate levels compared to the same routine performed under normoxic conditions, providing proof of principle that hypoxia heightens metabolite accumulation. Several studies show that hypoxia enhances the hypertrophic response to resistance training. Nishimura and colleagues (198) reported significantly greater increases in elbow flexor cross-sectional area when 4 sets of 10 repetitions at 70% of 1RM were performed under conditions of acute hypoxia versus normoxia. Similar findings have been reported elsewhere (19). Although these findings do not provide a causal link between hypertrophy and metabolic stress, they raise the possibility that increased metabolite accumulation plays a role in the process (247). It should be noted that reactive oxygen species (ROS) released during conditions of hypoxia have been implicated in carrying out a variety of signaling responses, providing another

(continued)

Blood Flow Restriction *(continued)*

potential explanatory mechanism for hypoxic-induced hypertrophy to heightened metabolic stress (19), possibly mediated by direct inducement of myogenesis via activation of HIF-1α (40). Moreover, not all studies have shown greater hypertrophy when training under hypoxic versus normoxic conditions (75), suggesting that differences in the methods by which hypoxia is delivered and the duration of exposure may cause different effects on muscular adaptations. How metabolic stress factors into the response remains speculative.

increases in hypertrophy associated with metabolic stress, it appears that other factors likely play a role as well. Suga and colleagues (261) demonstrated that only 31% of subjects displayed recruitment of fast-twitch fibers during occlusion training at 20% of 1RM compared with 70% of subjects who performed nonoccluded training at 65% of 1RM. Considering that BFR at this intensity (20% of 1RM) has been shown to increase muscle growth to an extent similar to, or greater than, high-intensity resistance training (150, 306), the anabolic effects seemingly cannot be solely a function of equal fiber recruitment. These findings are further supported by research showing significantly higher surface EMG amplitudes when traditional training is carried out at 80% of 1RM compared to occluded training at 20% of 1RM, indicating reduced muscle activation at the lower intensity (165). Recent studies investigating heavy- versus light-load training also show significantly greater muscle activation during the higher-intensity bout despite an apparently much greater metabolite accumulation during the light-load condition (5, 46, 242). However, surface EMG only provides insights into neural drive, which encompasses not only recruitment but also rate coding, synchronization, muscle fiber propagation velocity, and intracellular action potentials (16, 53).

Muddle and colleagues (191) decomposed EMG signals to provide more accurate insights into motor unit recruitment during low-load (30% maximal voluntary isometric contraction) versus high-load (70% maximal voluntary isometric contraction) training to muscular failure. Results showed that although fatigue brought about progressively greater recruitment of high-threshold motor units during light-load training, the extent of high-threshold recruitment was greater during heavier loading. More recently, Morton and colleagues (190) reported that high- and low-load training results in similar glycogen depletion in Type I and Type II fibers when sets are taken to momentary muscle failure; although metabolic stress was not measured, this suggests that the buildup of metabolites may have contributed to enhancing fiber recruitment. Importantly, while motor unit recruitment is required for muscle growth to occur, recruitment alone is not necessarily sufficient to promote hypertrophy; fibers also must be adequately stimulated to bring about an adaptive response.

Myokine Production

Metabolic stress may influence growth by upregulating anabolic myokines or downregulating catabolic myokines, or both (239). Although there is a logical basis for this claim, research on the topic is equivocal. Takarada and colleagues (265) demonstrated a gradual increase in IL-6 following multiple sets of knee extensions with BFR compared to volume-matched exercise without occlusion; levels remained elevated 24 hours post-exercise. The effect size was small, however, and the absolute amount of the increase was only 1/4 that reported for heavy-load eccentric exercise. Fujita and colleagues (80) found that 6 days of leg extensor occlusion training increased thigh cross-sectional area by 2.4% without any changes noted in IL-6 levels. Similarly, other studies showed that IL-6 levels remained unchanged following BFR training protocols known to elevate metabolic stress (1, 78). The totality of these findings seem to refute a role for IL-6 in hypertrophy induced by metabolic stress. The correlation between metabolic stress and other local growth factors has not been well studied, precluding the ability to draw conclusions regarding their potential relevance.

Evidence suggests that metabolic stress may influence muscle growth by downregulating local catabolic factors. Kawada and Ishii (129) reported significantly decreased MSTN levels in the plantaris muscle of Wistar rats following BFR exercise versus a sham-operated control group. Conversely, no differences in MSTN gene expression were seen in humans 3 hours after low-intensity exercise with and without occlusion (55). Another human trial showed that although BFR had no effect on MSTN, it downregulated several important proteolytic transcripts (FOXO3A, atrogin-1, and MuRF-1) 8 hours after exercise compared to a nonoccluded control group (166). In a study of physically active males, Laurentino and colleagues (150) investigated the effects of BFR on chronic MSTN levels following 8 weeks of training. Results showed a significant 45% reduction in MSTN gene expression with BFR compared to a nonsignificant reduction when performing low-intensity exercise without occlusion. The conflicting nature of these findings makes it difficult to formulate conclusions about whether hypertrophic adaptations from metabolic stress are related to alterations in myokine production. Moreover, none of the trials directly compared results to a heavy-load condition, further clouding the ability to draw causal inferences on the topic.

Cell Swelling

Another mechanism purported to mediate hypertrophy via metabolic stress is an increase in intracellular hydration (i.e., *cell swelling*). Cell swelling is thought to serve as a physiological regulator of cell function (107, 108). A large body of evidence demonstrates that an increase in the hydration status of a cell concomitantly increases protein synthesis and decreases protein breakdown. These findings have been shown in a wide variety of cell types, including osteocytes, breast cells, hepatocytes, and muscle fibers (147).

Current theory suggests that an increase in cellular hydration causes pressure against the cytoskeleton and cell membrane, which is perceived as a threat to the cell's integrity. In response, the cell upregulates an anabolic signaling cascade that ultimately leads to reinforcement of its ultrastructure (148, 240). Signaling appears to be mediated via integrin-associated volume osmosensors within cells (158). These sensors turn on anabolic protein-kinase transduction pathways, which are thought to be mediated by local growth factors (41, 146). PI3K appears to be an important signaling component in modulating amino acid transport in muscle as a result of increased cellular hydration (158). Research suggests that anabolic effects are also carried out in an mTOR-independent fashion (237), with evidence of direct regulation by MAPK modules (68, 236). Moreover, swelling of myofibers may trigger the proliferation of satellite cells and promote their fusion to the affected fibers (50), providing a further stimulus for growth.

Evidence is lacking as to whether cell swelling resulting from exercise-induced metabolic stress promotes hypertrophy. However, a sound rationale can be made for such an effect. Resistance exercise acutely alters intra- and extracellular water balance (253), and the extent of alterations depends on the type of exercise and the intensity of training. Cell swelling is thought to be heightened by resistance training that generates high amounts of lactic acid via the osmolytic properties of lactate (76, 252), although some research refutes this hypothesis (284). Intramuscular lactate accumulation activates volume regulatory mechanisms with effects seemingly amplified by the associated increased acidosis (147). Fast-twitch fibers are thought to be especially sensitive to osmotic changes, presumably because they contain high concentrations of aquaporin-4 (AQP4) water transport channels (76). Considering that fast-twitch fibers have been shown to have the greatest growth potential (134), an increased swelling in these fibers could conceivably enhance their adaptation in a meaningful way.

In an effort to determine whether cell swelling mediates hypertrophic adaptations, Gundermann and colleagues (98) carried out a randomized crossover study whereby 6 young males performed a low-intensity resistance exercise bout with BFR consisting of 4 sets of leg extensions at 20% of 1RM and a similar low-intensity resistance exercise bout without BFR; at least 3 weeks separated the trials. During the non-BFR session, a pharmacological vasodilator (sodium nitroprusside) was infused into the femoral artery immediately post-exercise to simulate the cell-swelling response induced by BFR exercise. Mixed-muscle protein synthesis measured 3 hours post-exercise showed increases only in the BFR condition; pharmacological vasodilation was insufficient to promote an anabolic response. While these findings appear to discount the role of cell swelling in hypertrophy, it must be noted that the blood flow response immediately post-exercise was almost twice as large in the BFR condition than in the non-BFR condition. Moreover, the non-BFR protocol involved performing the same number of repetitions at the same load as the BFR condition; thus, the intensity of effort in the non-BFR condition was very low, with sets stopped well short of muscular fatigue, resulting in suboptimal stimulation of myofibers. Taken together, it is difficult to draw relevant practical implications from the findings as to what, if any, role cell swelling plays in exercise-induced muscular adaptations.

Systemic Hormone Production

It has been posited that acute post-exercise elevations in anabolic hormones resulting from metabolite accumulation during resistance training may augment the hypertrophic response. In particular, exercise-induced metabolic stress is strongly associated with a spike in post-workout growth hormone levels (93-95, 102, 218, 264, 265). Although transient, the magnitude of these elevations is sizable. One study reported a 10-fold increase in GH levels with BFR training over and above that seen with similar-intensity nonoccluded exercise (81); another showed that post-workout increases reached 290-fold over baseline (265). Post-exercise elevations are thought to be mediated by a heightened accumulation of lactate or H+, or both (93, 102). People who lack *myophosphorylase*, a glycolytic enzyme responsible for breaking down glycogen and thus inducing lactate production, demonstrate an attenuated post-exercise growth hormone response (87), providing strong evidence for a link between lactate production and GH release. A metabolite-induced decrease in pH also may augment GH release via chemoreflex stimulation regulated by intramuscular metaboreceptors and group III and IV afferents (154, 291).

Given that GH is known to potentiate IGF-1 secretion, it seems logical that metabolite accumulation would be associated with increased post-exercise IGF-1 levels as well. This has been borne out to some extent by studies showing significantly greater IGF-1 elevations following the performance of metabolically fatiguing routines (135, 136, 232), although other research has failed to find such an association (138). Moreover, several (2, 81, 264), but not all (55), studies have reported acute increases in post-exercise IGF-1 levels following BFR training, which suggests that the results were mediated by metabolic stress. Importantly, the body of research is specific to the circulating IGF-1 isoform, and findings cannot necessarily be extrapolated to intramuscular effects.

The effect of metabolic stress on acute testosterone elevations remains unknown. Lu and colleagues (159) reported that exercise-induced lactate production correlated with increases in testosterone during a bout of high-intensity swimming in Sprague-Dawley rats. In a second component of the study, direct infusion of lactate into rat testes was found to cause a dose-dependent elevation in testosterone levels. On the other hand, controlled research in humans has produced disparate findings. While some studies show higher post-exercise testosterone release following metabolically fatiguing protocols compared with those that do not cause significant metabolite buildup (29, 95, 102, 174, 254), others show no significant differences

(135, 223, 260). In addition, a majority of BFR studies have failed to find significantly higher acute testosterone elevations despite high levels of metabolites (81, 223, 291), casting doubt as to whether the hormone is affected by metabolite accumulation. Inconsistencies between studies may be related to demographic factors such as sex, age, and training experience, and nutritional status also has been shown to affect testosterone release (139). As noted in chapter 1, whether transient post-exercise hormonal spikes have an effect on hypertrophic adaptations remains questionable. If there is such an effect, it would seem to be of small consequence and likely not a meaningful contributor to metabolite-induced anabolism.

Muscle Damage

Intense exercise, particularly when it is unaccustomed, can cause damage to skeletal muscle (45, 59, 144). This phenomenon, commonly known as *exercise-induced muscle damage* (EIMD), can be specific to just a few

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CONCLUSIONS ABOUT THE HYPERTROPHIC ROLE OF METABOLIC STRESS

Strong evidence suggests that exercise-induced metabolic stress contributes to the hypertrophic response. What remains to be determined is whether these effects are additive to the stimulus from mechanical forces or perhaps redundant provided a given loading threshold is achieved. The difficulty in trying to draw inferences from experimental resistance training designs is that mechanical tension and metabolic stress occur in tandem, confounding the ability to tease out the effects of one from the other. This can result in mistakenly attributing growth to metabolic factors when mechanical factors are, in fact, responsible, or vice versa.

The ability to draw a cause–effect relationship between metabolic stress and hypertrophy is further confounded by the fact that the exercise-induced buildup of metabolites generally occurs in tandem with damage to myofibers. Given the commonly held belief that damaging exercise mediates anabolism (241), it is difficult to distinguish the effects of one variable from the other with respect to hypertrophic adaptations. Research showing that blood flow restriction training increases muscle growth without significant damage to fibers suggests that the hypertrophic effects of metabolite accumulation are indeed separate from myodamage (157), although conflicting evidence on the topic renders a definitive conclusion premature (299). Some evidence indicates that metabolic stress only contributes to muscle growth during performance of low-load resistance training, with no additive hypertrophic effects seen when training with heavy loads (80% of 1RM), at least at the whole-muscle level (18).

Finally and importantly, the mechanisms responsible for anabolic effects of metabolic stress have not been fully elucidated. Although it is conceivable that increased muscle fiber recruitment is the primary mechanism by which metabolite accumulation induces hypertrophic adaptations, it seems unlikely that this phenomenon solely accounts for any or all of the observed effects. Rather, evidence suggests that the combined integration of multiple local and perhaps systemic factors contributes to muscular growth in a direct or permissive manner, or both (302). The fact that human studies to date have been primarily carried out in untrained subjects leaves open the prospect that mechanisms may differ based on training experience.

macromolecules of tissue or manifest as large tears in the sarcolemma, basal lamina, and supportive connective tissue, as well as injury to contractile elements and the cytoskeleton (figure 2.6) (289). EIMD generally sets off a cascade of subsequent responses that include local inflammation, disturbed Ca^{2+} regulation, activation of protein breakdown, and secretion of substances from damaged fibers that result in increased blood-levels of proteins such as creatine kinase (294). The severity of EIMD depends on factors such as the type, intensity, and total duration of training (164). In a recent review, Hyldahl and Hubal (121) proposed that EIMD exists on a continuum spanning from favorable adaptive cell signaling with mild damage to maladaptive responses such as pervasive membrane damage and tissue necrosis with severe myocellular disruptions.

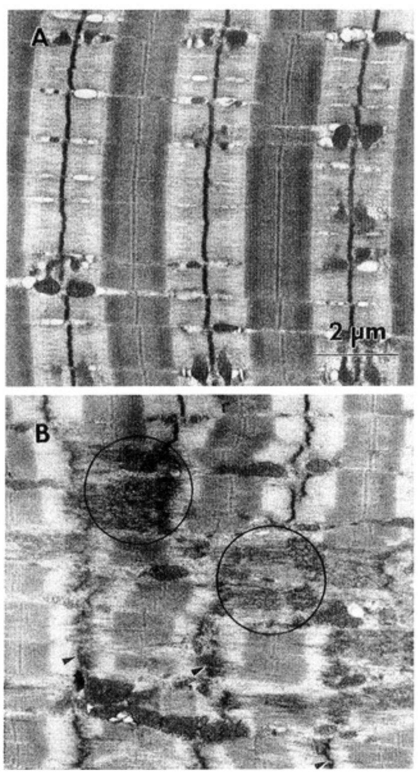

FIGURE 2.6 Sarcomere disruption following eccentric contractions. *(a)* Sarcomeres from normal muscle show excellent alignment and regular banding patterns; *(b)* sarcomeres from muscle exposed to eccentric contractions show regions of Z-disc streaming and frank sarcomere disruption next to sarcomeres that appear normal.

Reprinted by permission from R.L. Lieber, T.M. Woodburn, and J. Friden, "Muscle Damage Induced by Eccentric Contractions of 25% Strain," *Journal of Applied Physiology* 70, no. 6 (1991): 2498-2507.

EIMD is highly influenced by the type of muscular action. Although concentric and isometric exercise can bring about EIMD, eccentric actions have by far the greatest impact on its manifestation (43, 83). Eccentrically induced EIMD is more prevalent in fast-twitch than in slow-twitch fibers (290). Possible reasons include a reduced oxidative capacity, higher levels of tension generated during training, and structural differences between fiber phenotypes (220).

Damage from eccentric actions are attributed to mechanical disruption of the actomyosin bonds rather than ATP-dependent detachment, thereby placing a greater strain on the involved machinery in comparison to concentric and isometric actions (62). Studies show that the weakest sarcomeres reside in different aspects of each myofibril, leading to speculation that the associated nonuniform lengthening results in a shearing of myofibrils. This sets off a chain of events beginning with a deformation of T-tubules and a corresponding disruption of calcium homeostasis that mediates the secretion of the calcium-activated neutral proteases (such as calpain) involved in further degradation of structural muscle proteins (6, 17). There is evidence of a dose–response relationship, whereby higher exercise volumes correlate with a greater degree of myodamage (199). Symptoms of EIMD include decreased force-producing capacity, increased musculoskeletal stiffness and swelling, delayed-onset muscle soreness (DOMS), and a heightened physiological stress response typified by an elevated heart rate response to submaximal exercise and heightened lactate production (270).

EIMD decreases when a person performs the same exercise program on a consistent basis; a phenomenon commonly known as the *repeated bout effect* (177). Several factors are thought to be responsible for this effect, including an adaptive strengthening of connective tissue, increased efficiency in the recruitment of motor units, enhanced synchronization of motor units, a more even distribution of the workload among fibers, and a greater contribution of muscle synergists (24, 270). Research indicates that α7β1 integrin may be involved in the process. It has been shown that α7β1 integrin expression is increased after a damaging exercise bout, which then initiates transcription of genes that afford protection from future mechanical stress as well as potentially promoting anabolism (163).

Effects of the repeated bout effect are seen rapidly after unaccustomed exercise. Evidence suggests that just one additional bout of the same exercise protocol reduces the EIMD-associated swelling response to just 1/3 of the initial bout (70). Similarly, Chen and colleagues (36) reported marked reductions in damage-related markers (soreness, creatine kinase activity, and plasma myoglobin concentration) in a follow-up bout of eccentric exercise performed 2 weeks after a bout consisting of the same training protocol. Repeated training with the same exercise program over time further diminishes EIMD-associated effects, as elegantly demonstrated in a study by Damas and colleagues (49), who tracked indices of myodamage across 10 weeks of regimented resistance training carried out to volitional muscular failure. Results showed substantial EIMD after the initial training session; however, damage was markedly attenuated by the fifth session and practically inconsequential 48 hours after the last session. Other research has shown a similar attenuation of damage-related markers during longitudinal training programs when performing the same routine over time (18). The effects of the repeated bout effect can last for several months, even in the absence of unaccustomed training during this period. Evidence that the upper-extremity muscles have a greater predisposition to EIMD than the leg muscles suggests a protective benefit in muscles that are frequently used during everyday activities (37). Other factors that can influence the magnitude of the protective effect include the specific composition of exercise variables (e.g., training intensity, velocity, number of damaging contractions), muscle length, muscle group, age, and sex (122).

Although EIMD can be deleterious from a performance standpoint, some researchers have speculated that the associated increases in inflammation and protein turnover are necessary for muscle growth (63, 300). The rationale is based on the hypothesis that structural alterations associated with damage influence gene expression in a manner that strengthens the affected tissue, thereby serving to protect the muscle against further injury (12). Substantial evidence links muscle damage to factors involved in the hypertrophic response to exercise, although these correlations do not establish causality for a positive effect.

Despite the sound theoretical basis, there is a dearth of research directly investigating the causal relationship between EIMD and muscle growth. Exposure of murine tibialis anterior muscles to injury-producing myotoxin injection was found to result in both larger muscle fibers and a 3-fold greater satellite count compared to uninjured fibers (105). Moreover, evidence shows transplantation of satellite cells into damaged myofibers elicits an increase in muscle hypertrophy across an animal's life span (103). Taken together, these data suggest that myocellular damage alone, as well as in combination with an increase in the number of satellite cells, can provide a sufficient stimulus to elicit muscle growth. However, the protocols employed have minimal relevance to human exercise protocols, and thus the practical implications of these findings are limited.

Alternatively, Komulainen and colleagues (132) exposed the tibialis anterior muscles of anesthetized Wistar rats to repeated concentric or eccentric muscle actions. The eccentric muscle actions produced massive injury to the muscle; beta-glucuronidase activity (a measure of myodamage) showed a 7.1-fold increase from baseline. Alternatively, concentric muscle actions resulted in a modest 2.6-fold increase in beta-glucuronidase activity, indicating that the damage was relatively minor. Similar increases in muscle cross-sectional area were noted in both groups, suggesting a threshold for EIMD-induced growth beyond which myodamage provides no additional beneficial hypertrophic effects. The study is confounded by evaluating polar-extreme levels of damage. Whether a dose–response relationship exists between hypertrophy and moderate levels of EIMD, therefore, cannot be determined. Moreover, the severe damage experienced in

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CHALLENGES TO THE EIMD HYPOTHESIS

As discussed, muscles become increasingly less susceptible to damage with recurring exercise—a function of the repeated bout effect. This phenomenon would seem to rule out potential involvement of EIMD in the hypertrophic response of those who are well trained (199). However, evidence suggests that myodamage is indeed present in trained lifters when performing unaccustomed exercise, albeit to a lesser extent than in novices. Gibala and colleagues (84) recruited six resistance-trained men to perform 8 sets of 8 repetitions at a load equivalent to 80% of 1RM. The researchers employed a unilateral protocol whereby one arm performed only concentric actions while the other arm performed only eccentric actions. Muscle biopsies taken 21 hours after the exercise bout showed a significantly greater disruption in fibers from the eccentrically trained arms than in the concentrically trained arms. These findings underscore the fact that the repeated bout effect only attenuates the magnitude of muscle damage on a general level as opposed to preventing its occurrence when a novel routine is employed and leaves open the possibility that EIMD may contribute to hypertrophy in well-trained lifters. It seems that the key to this response is to provide a novel stimulus by altering training variables in an unaccustomed fashion.

Some researchers have questioned whether EIMD confers any anabolic effects, based on research showing marked hypertrophy from low-intensity BFR training with ostensibly minimal tissue damage (2, 265). The BFR technique combines light loads (20% to 50% of 1RM) with occlusion via pressure cuff to impede venous return without obstructing arterial inflow. Regular performance of BFR induces marked hypertrophy, often similar to what is observed with the use of heavy loads. Given the light loads employed, it is hypothesized that BFR confers these hypertrophic benefits while minimizing disruption of myofibers. However, muscle damage is a known consequence of reperfusion subsequent to ischemia (72, 99). Takarada and colleagues (265) demonstrated that although markers of muscle damage were attenuated after BFR training, there was evidence of fine microdamage within myofibers, leaving open the possibility that damage may have contributed to the results. Moreover, it remains possible that hypertrophy would have been enhanced to an even greater extent had EIMD been heightened in the BFR group. Markers of muscle damage following BFR have been demonstrated elsewhere, including lengthy decrements in maximal voluntary contraction, heightened delayed-onset muscle soreness, and elevated sarcolemmal permeability (64, 298, 299).

Some investigators have questioned whether EIMD mediates hypertrophic adaptations based on research showing that downhill running can induce significant damage to muscle tissue without corresponding growth (24). This observation, however, fails to take into account the unique molecular responses associated with aerobic versus resistance exercise, and the corresponding post-workout stimulation of myofibers. The two types of training activate and suppress distinctly different subsets of genes and cellular signaling pathways (109), thereby bringing about divergent muscular adaptations. It also should be noted that damage elicited by aerobic training manifests differently from that elicited by resistance exercise. Peak creatine kinase activity is noted approximately 12 to 24 hours after downhill running, whereas that associated with resistance training is not evident until about 48 hours after the training bout and can peak 4 to 6 days post-workout (246). In addition, downhill running is associated with peak creatine kinase levels of between 100 to 600 IU, whereas those of resistance range from 2,000 to 10,000 IU (44). The implications of these variances remain to be established. Moreover, creatine kinase levels do not necessarily reflect the degree or time course of myodamage (45), calling into question their practical relevance with respect to exercise training. What can be inferred from aerobic training data is that muscle damage by itself is not sufficient to induce significant muscle growth. Thus, if EIMD does play a role in the hypertrophic response to exercise, it can do so only in the presence of resistance-based mechanical overload.

the eccentric muscle actions may have been so excessive that it negatively affected remodeling.

In a human trial on the topic, Flann and colleagues (69) randomly assigned 14 young, healthy men and women into one of two groups: (1) a control group that engaged in eccentric cycle ergometry at a "somewhat hard" level (gauged by a rating of perceived exertion scale; training was performed 3 times per week for 20 minutes over an 8-week period), and (2) a pretrained group that carried out a protocol identical to the control group's, except that it included a 3-week ramp-up period during which subjects performed exercise at a low intensity to gradually acclimate their muscles to the training stimulus. At the study's end, similar increases in muscle girth were found between the groups. Although these results are intriguing, the study had numerous methodological limitations, including the use of untrained subjects, unequal training duration between the groups, and a small sample size that compromised statistical power. In addition, the pretrained group showed evidence of myodamage as assessed by elevated creatine kinase levels, although the extent was significantly less than that noted in the control group. This raises the possibility that the magnitude of damage sustained by those who were pretrained was adequate to maximize any added hypertrophic adaptations. Alternatively, it remains conceivable that EIMD incurred during training by the untrained subjects exceeded the body's reparative capabilities, ultimately mitigating growth by impairing the ability to train with proper intensity and delaying supercompensatory adaptations.

A recent study by Damas and colleagues (49) has been interpreted by some as refuting a hypertrophic role of EIMD. In the study, 10 untrained young men performed a 10-week progressive resistance training program consisting of the leg press and leg extension exercises (3 sets of each exercise, 9RM to 12RM per set, 90-second rest between sets) carried out twice weekly. Myofibrillar protein synthesis and muscle damage were assessed after the first training session, after 3 weeks of training, and at the end of the 10-week study period. Results showed greater increases in protein synthesis after the initial exercise bout than after later bouts. Muscle damage, as determined by Z-band streaming, was greatest after the initial bout as well, and rapidly declined to minimal levels by the end of the study. Most interestingly, despite the high correlation between the initial damaging bout and high levels of protein synthesis, these outcomes did not correlate with muscle growth obtained at the conclusion of the study period; only after attenuation of EIMD at week 3 did results show an association between protein synthesis and hypertrophy. This has led to speculation that exercise-induced muscle protein synthesis is only directed to generating muscle hypertrophy after attenuation of EIMD.

However, such conclusions appear to be an overextrapolation of the findings by Damas and colleagues (49). While the study elegantly demonstrated that an initial bout of damage was explanatory as to why muscle protein synthesis is not necessarily associated with exercise-induced hypertrophy over time, the data cannot be used to draw inferences about long-term effects of damage on muscular adaptations. To properly study the topic would require carrying out a longitudinal resistance training study in which one group experiences mild to moderate damage and then compare those results with another group that experiences minimal damage. Unfortunately, such a design is problematic because attempting to isolate EIMD in this fashion would involve altering other resistance training variables that would confound the ability to determine causality. It is impossible to determine whether some level of muscle damage experienced by subjects in the study by Damas and colleagues (49) contributed to the observed hypertrophic changes. Moreover, it is not clear whether more (or less) damage may have influenced hypertrophy over time. The only conclusion that can be made is that the damage experienced in an initial exercise bout in untrained individuals appears to be directed toward structural repair as opposed to hypertrophy; the effects of repeated exposure to varying levels of damage beyond the initial bout cannot be deduced from the study design. Overall, the difficulty in controlling confounding variables when attempting to study the effects of EIMD on hypertrophy in human trials precludes our ability to draw relevant inferences.

The regeneration and repair of muscle tissue following EIMD is carried out by novel transcriptional programs that are associated with or promoted by inflammatory processes, satellite cell activity, IGF-1 production, and cell swelling (162). Following is an overview of factors hypothesized to promote an EIMD-induced hypertrophic response.

Inflammatory Processes

The body's response to EIMD can be equated to its response to infection (240). After a damaging exercise bout, neutrophils migrate to the injury site while agents are released by affected fibers that attract macrophages to the region as well (176). This sets off a cascade of events in which inflammatory cells then secrete other substances to facilitate the repair and regeneration of damaged muscle. Inflammatory processes resulting from EIMD can have either

KEY POINT

Research remains equivocal as to whether EIMD can enhance muscular adaptations, and excessive damage most certainly has a negative effect on muscle development. If EIMD does in fact mediate muscular adaptations, it remains to be determined the extent to which these proposed mechanisms are synergistic and whether an optimal combination exists to maximize the hypertrophic response to resistance training.

a beneficial or deleterious effect on muscular function depending on the magnitude of the response, previous exposure to the applied stimulus, and injury-specific interactions between the muscle and inflammatory cells (277).

Neutrophils are more abundant in the human body than any other type of white blood cell. In addition to possessing phagocytic capabilities, neutrophils release proteases that aid in breaking down cellular debris from EIMD. They also secrete cytolytic and cytotoxic substances that can exacerbate damage to injured muscle and inflict damage to healthy neighboring tissues (277). Hence, their primary role in skeletal muscle is likely confined to myolysis and other facets associated with the removal of cellular debris as opposed to the regeneration of contractile tissue.

Despite a lack of evidence directly linking neutrophils to hypertrophy, it is conceivable that they may mediate anabolism by signaling other inflammatory cells necessary for subsequent muscle remodeling. One such possibility is reactive oxygen species (ROS) (283), which have been shown to mediate intracellular signaling in response to intense physical activity (89, 126, 127, 217, 273). Neutrophils are associated with the production of numerous ROS variants, including hydrogen peroxide, superoxide, hydroxyl radical, and hypochlorous acid (131). ROS are associated with hypertrophy of both smooth muscle and cardiac muscle (262), and some speculate that anabolic effects extend to skeletal muscle as well (265). In support of this hypothesis, transgenic mice displaying suppressed levels of selenoproteins (a class of proteins that act as powerful antioxidants) had 50% more muscle mass following synergist ablation compared to wild-type controls (112). Moreover, supplementation with antioxidants that inhibit ROS production impairs both intracellular anabolic signaling and muscle hypertrophy following muscular overload (181). Collectively, these findings suggest that redox-sensitive signaling pathways may enhance exercise-induced muscular adaptations.

ROS have been shown to mediate anabolism via activation of the MAPK pathway. The treatment of C2 myoblasts with an ROS variant heightens MAPK signaling, and the temporal response varies between MAPK subfamilies (ERK1/2, JNK, and p38 MAPK) (130). Given that eccentric exercise is associated with greater MAPK activation compared to concentric or isometric actions (162, 167), it is conceivable that ROS production contributes to this stimulus. There also is evidence that ROS enhance growth processes by amplifying IGF-1 signaling. In vitro ROS treatment of mouse C2C12 myocytes significantly increased phosphorylation of the IGF-1 receptor, whereas phosphorylation was markedly suppressed with antioxidant provision (104). These findings suggest a crucial role for ROS in the biological actions of IGF-1.

Interestingly, there is evidence that ROS interfere with the signaling of various serine/threonine phosphatases, such as calcineurin. ROS activity impairs calcineurin activation by blocking its calmodulin-binding domain (33). Calcineurin is thought to be involved in both skeletal muscle growth (58, 183) and fiber phenotype transformation (209), and thus its inhibition may be detrimental to anabolism. Moreover, some studies have failed to demonstrate that ROS are in fact activated in response to EIMD (235). When considering the body of literature as a whole, any anabolic effects of ROS are likely dependent on exercise mode (i.e., anaerobic versus aerobic), the species of ROS produced, and perhaps other factors.

In contrast to neutrophils, research indicates a potential role for macrophages in regenerative processes following EIMD (277), and some researchers even speculate that they are necessary for muscle growth (131). Macrophages appear to exert anabolic effects by secreting local growth factors associated with inflammatory processes. Vascular endothelial growth factor (VEGF), a myokine considered crucial for overload-induced hypertrophy, is thought to play a particularly important role in the process. Following muscle injury, VEGF acts as a chemoattractant for macrophages, initiating the inflammatory response and release of IGF-1, among other anabolic agents (120). Macrophage accumulation is associated with an increased satellite cell content, providing another potential mechanism for enhancing muscle growth (295).

It was originally thought that myodamage directly led to the production of pro-inflammatory myokines (26, 213). Although this would seem to have a logical basis, more recent research indicates that such myokine production may be largely independent of EIMD. A study by Toft and colleagues (278) showed that IL-6 levels were only modestly elevated relative to increases in creatine kinase following 60 minutes of eccentric cycle ergometry exercise, suggesting a weak association between EIMD and IL-6 production. These results are consistent with those of other studies showing poor correlation in the time course of IL-6 and creatine kinase appearance (47). The totality of findings has led to the supposition that IL-6 release is predominantly a function of muscle contraction. Mechanistically, some researchers have hypothesized that this facilitates the mobilization of substrate from fuel depots so that glucose homeostasis is maintained during intense exercise (65).

It is important to note that only IL-6 and IL-8 have been shown to be released from skeletal muscle in the absence of damaging exercise (35). Many other myokines may play a role in the hypertrophic response to EIMD. Systemic IL-15 levels and IL-15 mRNA in skeletal muscle are markedly elevated after eccentric (but not concentric) exercise, giving credence to the notion that elevations are contingent on damage to fibers (27, 224). Some studies show that IL-15 directly regulates hypertrophy by increasing muscle protein synthesis and reducing proteolysis in differentiated myotubes (197, 221), although these findings recently have been challenged (219). There also is evidence that fibroblast growth factors (FGFs)—powerful proliferative agents involved in hypertrophic processes—are preferentially upregulated following eccentric exercise. Research indicates that FGFs are secreted from damaged fibers (41) and that their time course of release parallels the increased creatine kinase levels associated with EIMD (42). These findings lend mechanistic support to the hypothesis that damaging exercise promotes an anabolic stimulus.

Satellite Cell Activity

A large body of evidence links EIMD with satellite cell activity (52, 233, 245). Damaged myofibers must rapidly acquire additional myonuclei to aid in tissue repair and regeneration or otherwise face cell death. Satellite cells facilitate these means by proliferating and fusing to damaged fibers. Because satellite cells tend to populate under the myoneural junction (111, 251), it is speculated that they may be further stimulated by activation from motor neurons innervating damaged fibers, enhancing the regenerative response (289). It has been hypothesized that under certain conditions, stimulated satellite cells fuse to each other to form new myofibers (13), but evidence as to how this relates to traditional resistance training practices is lacking.

Initial signaling to activate satellite cells following EIMD is purported to originate from muscle-derived nitric oxide, potentially in combination with the release of HGF (4, 269, 277). The process appears to be controlled at least to some extent by the cyclooxygenase (COX)-2 pathway, which is considered necessary for maximizing exercise-induced hypertrophic adaptations (257). COX-2 acts to promote the synthesis of prostaglandins believed to stimulate satellite cell proliferation, differentiation, and fusion (22). Research shows an enhanced myogenic response when inflammatory cells are abundant and a blunted response in their absence (22), suggesting that inflammatory processes subsequent to damaging exercise are critical to remodeling. The hypertrophic importance of COX-2 is further supported by research investigating the effects of COX-inhibiting nonsteroidal anti-inflammatory drugs (NSAIDs) on the post-exercise satellite cell response (11). A majority of studies show decreased post-exercise satellite cell activity when NSAIDs are administered (22, 23, 161, 184), which would conceivably limit long-term muscle growth, although these findings are not universal (212). It is important to point out that mechanical stimuli alone can instigate satellite cell proliferation and differentiation even without appreciable damage to skeletal muscle (196, 296). Hence, it is not clear whether the effects of EIMD are additive or redundant with respect to maximizing muscle protein accretion.

IGF-1 Production

There is evidence that EIMD potentiates IGF-1 production and, thus, given the anabolic functions of this hormone, may enhance muscle growth. McKay and colleagues (178) studied the effects of performing a series of 300 lengthening knee extension contractions on all three IGF-1 isoforms in untrained young men. The results showed a significant increase in MGF mRNA 24 hours post-exercise. Intriguingly, expression of both IGF-1Ea and IGF-1Eb mRNA were not elevated until 72 hours after training. The early-phase activation of MGF as a result of the damaging protocol suggests that this IGF-1 isoform is preferentially involved in the repair and remodeling process following EIMD. Similarly, Bamman and colleagues (10) evaluated the effects of 8 sets of 8 eccentric versus concentric actions on muscle IGF-1 mRNA concentration. The eccentric exercise resulted in a significant 62% increase in IGF-1 mRNA concentrations as opposed to a nonsignificant increase from concentric exercise. In addition, eccentric exercise caused a reduction of IGFBP-4 mRNA—a strong inhibitor of IGF-1—by 57%, whereas the concentric condition showed only modest changes in the levels of this protein. Importantly, results were positively correlated with markers of muscle damage, suggesting that the IGF-1 system is involved in the repair process.

The association between EIMD and IGF-1 upregulation has not been universally confirmed in the literature. Garma and colleagues (82) compared the acute anabolic response of volume-equated bouts of eccentric, concen-

PRACTICAL APPLICATIONS

EFFECT OF NSAIDS ON MUSCLE HYPERTROPHY

Nonsteroidal anti-inflammatory drugs (NSAIDs) are a class of analgesics commonly used to relieve the pain and swelling associated with delayed-onset muscle soreness. NSAIDs are thought to promote pain-reducing effects by inhibiting the activity of cyclooxygenase (COX), a family of enzymes that catalyze the conversion of arachidonic acid to pro-inflammatory prostanoids (30, 288). Thirty million people are estimated to take NSAIDS daily (15), and their use is especially widespread among those who participate in intense exercise programs (297).

An often-overlooked issue with NSAID consumption in combination with resistance training, however, is the potential interference with muscular adaptations. In addition to the effects of NSAIDs on pain sensation, prostanoids also are purported to stimulate the upstream regulators of protein synthesis, including PI3K and extracellular signal-regulated kinases (79, 208, 228). Moreover, there is evidence that prostanoids are intricately involved in enhancing satellite cell proliferation, differentiation, and fusion (21), thereby facilitating greater muscle protein accretion (115). These data provide compelling evidence that COX enzymes are important and perhaps even necessary for maximizing resistance training–induced muscle hypertrophy (256).

Despite a seemingly logical basis for COX-mediated hypertrophic effects, acute studies on the use of NSAIDs do not seem to show a detrimental impact on post-exercise protein synthesis. Although NSAID administration in animal trials following chronic overload has consistently found impairments in protein metabolism (208, 228, 287), only one human trial showed a blunting of protein synthesis (280), whereas several others failed to note a deleterious effect (28, 185, 215). Discrepancies between findings may be related to methodological variances, physiological differences between species, or differences in the mechanisms of the NSAIDs used (i.e., selective versus nonselective COX inhibitors).

On the other hand, the body of literature strongly suggests that NSAID use interferes with satellite cell function. This has been shown in vitro (180, 207) as well as in vivo in both animal (21, 23) and human trials (161, 184). It has been proposed that hypertrophy is limited by a myonuclear domain ceiling, estimated at approximately 2,000 μm^2; beyond this ceiling, additional nuclei must be derived from satellite cells for further increases in hypertrophy to occur (216). Although a rigid ceiling threshold has been challenged (192), the general consensus among researchers is that when fibers reach a certain critical size, satellite cell–derived nuclei are needed to promote additional gains. Therefore, a blunting of satellite cell function would seemingly limit a person's hypertrophic potential by restricting the satellite cell pool.

How the acute data play out over the long term is not clear. Results from studies that have directly investigated the effects of NSAIDs on hypertrophy are conflicting. Consistent with the research on protein synthesis, animal studies indicate that NSAID administration markedly reduces overload-induced muscle growth (23, 201, 256). Alternatively, several human trials have either failed to demonstrate hypertrophic impairments (142, 214) or showed a positive effect from NSAID use during regimented resistance training (281). When attempting to reconcile differences between studies, it is possible that NSAIDs reduce protein breakdown to a similar or even greater degree than they suppress protein synthesis, thus resulting in a non-negative protein balance. Rodent studies lend support to this hypothesis (228). The fact that the study showing increased hypertrophy from the use of NSAIDs (281)

(continued)

Effect of NSAIDs on Muscle Hypertrophy *(continued)*

was carried out in elderly adults (60 to 85 years of age) raises the possibility that positive benefits were due to a suppression of chronic inflammation, which has been shown to impair anabolism and accelerate proteolysis (225). It is also conceivable that the extent of hypertrophy in human trials was below the subjects' myonuclear domain ceiling. This would seemingly explain why animal models using techniques designed to promote extreme rates of hypertrophy (i.e., synergist ablation, chronic stretch) far beyond that of traditional resistance training in humans show substantial hypertrophic impairment, because a robust satellite cell pool would be required for continued muscle growth.

Lilja and colleagues (153) conducted the most comprehensive human study to date on the topic in resistance-trained, young, healthy individuals. As opposed to previous work in young subjects (142), the study administered a higher dosage (1,200 mg/day vs. 400 mg/day of ibuprofen), and training was carried out over a longer study period (8 weeks vs. 6 weeks), thereby providing better insights into effects on muscular adaptations. Contrary to other findings, results showed that quadriceps growth in the NSAID group was blunted to approximately half that of the group receiving a placebo (3.7% vs. 7.5%, respectively). The researchers noted a downregulation of IL-6 in the NSAID group compared to the placebo group, providing a potential mechanistic explanation for the impaired gains associated with ibuprofen consumption. In summary, evidence indicates that the occasional use of NSAIDs does not impair muscle hypertrophy. Whether chronic NSAID administration is detrimental to muscle growth remains undetermined and likely is population specific: those with chronic low-grade inflammation might benefit from their use, whereas healthy, well-trained people could see long-term impairments.

tric, and isometric exercise in rodents. Results showed similar effects on cell signaling independent of the type of muscle action; no significant differences were seen in IGF-1 mRNA levels, and pre- to post-exercise increases were actually greatest in the isometric condition. The reason for these conflicting findings is not readily evident and likely relates to methodological differences in the studies.

Cell Swelling

As discussed earlier in the section on metabolic stress, cell swelling has been shown to positively regulate anabolic and anticatabolic processes. Specifically, increases in cellular hydration are associated with an increase in muscle protein synthesis and a concomitant decrease in proteolysis. The inflammatory response that accompanies damaging exercise involves a buildup of fluid and plasma proteins within damaged muscle. Depending on the extent of damage, the amount of accumulated fluid can exceed the capacity of the lymphatic drainage system, which leads to tissue swelling (100, 176, 220). Trauma to capillaries may increase the magnitude of edema (45). Swelling associated with an acute bout of eccentric elbow flexion exercise in untrained subjects produced an increase in arm circumference of as much as 9%, and values remained elevated for up to 9 days (118). Similarly, Nosaka and Clarkson (200) found that edema increased arm circumference by as much as 1.7 inches (4.3 cm) after unaccustomed eccentric exercise, and swelling was evident in all subjects by 3 days after performance. Although swelling is diminished over time with regimented exercise via the repeated bout effect, substantial edema can persist even in well-trained subjects for at least 48 hours post-workout provided a novel exercise stimulus is applied (117).

Whether the swelling associated with EIMD contributes to myofiber hypertrophy is unknown. Confounding factors make this an extremely difficult topic to study directly. There is some evidence that the use of NSAIDs, which blunt the inflammatory response and hence moderate the extent of cell swelling, impairs the increase in muscle protein synthesis normally associated with resistance exercise (208, 228, 280). It is feasible that deleterious effects on anabolism may be related to a decrease in cell swelling. However, these findings do not imply a cause–effect relationship between increased cellular hydration and muscle protein accretion; factors such as impaired satellite cell and macrophage activity may also be responsible for any negative effects. Moreover, other studies have failed to show an impaired muscle protein synthetic response following NSAID administration (28, 185), further clouding the ability to draw conclusions on the topic.

PRACTICAL APPLICATIONS

CONCLUSIONS ABOUT THE HYPERTROPHIC ROLE OF MUSCLE DAMAGE

A sound theoretical rationale exists for how EIMD may contribute to the accretion of muscle proteins. Although exercise-induced hypertrophy can apparently occur without significant myodamage (304), evidence implies that microtrauma enhances the adaptive response or at least initiates the signaling pathways that mediate anabolism. That said, a cause–effect relationship between EIMD and hypertrophy has yet to be established, and if such a relationship does exist, the degree of damage necessary to maximize muscle growth remains to be determined. Research does suggest a threshold for a hypertrophic stimulus, beyond which additional myodamage confers no further benefits and may in fact interfere with growth-related processes. There is clear evidence that excessive EIMD reduces a muscle's force-producing capacity. This in turn interferes with the ability to train at a high level, which impedes muscle development. Moreover, although training in the early recovery phase of EIMD does not seem to exacerbate muscle damage, it may interfere with the recovery process (143, 199). Taken as a whole, it can be speculated that a protocol that elicits a mild to moderate amount of damage may help to maximize the hypertrophic response. Considering that a ceiling effect slows the rate of hypertrophy as one gains training experience, EIMD may be particularly relevant to the anabolic response in well-trained people. These hypotheses require further study to ascertain their veracity.

TAKE-HOME POINTS

- Numerous intracellular signaling pathways have been identified in skeletal muscle including PI3K/Akt, MAPK, phosphatidic acid, AMPK, and calcium-dependent pathways. The serine/threonine kinase mTOR has been shown to be critical to mechanically induced hypertrophic adaptations.
- Mechanical tension is clearly the most important mechanistic factor in training-induced muscle hypertrophy. Mechanosensors are sensitive to both the magnitude and duration of loading, and these stimuli can directly mediate intracellular signaling to bring about hypertrophic adaptations.
- There is compelling evidence that the metabolic stress associated with resistance training promotes increases in muscle protein accretion. Hypothesized factors involved in the process include increased fiber recruitment, heightened myokine production, cell swelling, and systemic hormonal alterations. Whether the hypertrophic effects of metabolite accumulation are additive or redundant to mechanical tension is yet to be established.
- Research suggests that EIMD may enhance muscular adaptations, although excessive damage clearly has a negative effect on muscle development. Hypothesized factors involved in the process include the initiation of inflammatory processes, increased satellite cell activity, the mediation of IGF-1 production, and cell swelling. The extent to which these mechanisms are synergistic or redundant, and whether an optimal combination exists to maximize the hypertrophic response to resistance training, remains to be determined.